

1. (a) Definitions:

- $\Omega \neq \emptyset, \mathcal{C} \subset \mathcal{P}(\Omega)$ is semi-algebra if
 - (i) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}(\pi\text{-system})$;
 - (ii) $\forall A \in \mathcal{C} \Rightarrow A^C = \bigcup_{i=1}^k B_i, \{B_i\}_{i=1}^k \subset \mathcal{C}$ are disjoint;
- $\Omega \neq \emptyset, \mathcal{F} \subset \mathcal{P}(\Omega)$ is an algebra if
 - (i) $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$;
 - (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
 - (iii) $\Omega \in \mathcal{F}$.
- $\Omega \neq \emptyset, \mathcal{F} \subset \mathcal{P}(\Omega)$ is an σ -algebra if
 - (i) $\{A_i\} \subset \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$;
 - (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
 - (iii) $\Omega \in \mathcal{F}$.

(b) Will first show $A \Delta B \subset (A \Delta B) \cup (B \Delta C)$ and then the rest follows from the finite subadditivity of μ .

Choose $x \in A \Delta C = (A \cap C^c) \cup (A^c \cap C)$. We will deal with the case when $x \in (A \cap C^c)$ and show that x belongs to the right side (similar steps will prove the case when $x \in (A^c \cap C)$). So we have x in A but not in C :

If $x \in B$: then $x \in B \cap C^c \subset B \Delta C \subset (A \Delta B) \cup (B \Delta C)$.

If $x \in B^c$: then $x \in A \cap B^c \subset A \Delta B \subset (A \Delta B) \cup (B \Delta C)$.

2. (a) Taking $F = \Omega \in \mathcal{F}$, we get $\Omega_0 = \Omega \cap \Omega_0 \in \mathcal{F}_0$.

If $A = F \cap \Omega_0 \in \mathcal{F}_0$ for some $F \in \mathcal{F}$, then $\Omega_0 \setminus A = F^c \cap \Omega_0 \in \mathcal{F}_0$, since $F^c \in \mathcal{F}$ (note that $\Omega_0 \setminus A$ is the complement of A in Ω_0).

If $\{A_i\} \subset \mathcal{F}_0$, then each $A_i = F_i \cap \Omega_0$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\bigcup_{i \geq 1} F_i \in \mathcal{F}$ and hence $\bigcup_{i \geq 1} A_i = (\bigcup_{i \geq 1} F_i) \cap \Omega_0 \in \mathcal{F}_0$. This proves that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

(b) Take any $E \in \mathcal{P}(\Omega)$, we will show: $\mu^*(E) = \mu^*(A \cap E) + \mu^*(A^c \cap E)$. (0)

By sub-additivity of μ^* , we have: $\mu^*(E) \leq \mu^*(A \cap E) + \mu^*(A^c \cap E)$. (1)

Using monotonicity of μ^* , we get $0 \leq \mu^*(A \cap E) \leq \mu^*(A) = 0$. So, $\mu^*(A \cap E) = 0$. Using this and monotonicity of μ^* again, we have

$$\mu^*(E) \geq \mu^*(A^c \cap E) = \mu^*(A \cap E) + \mu^*(A^c \cap E) \quad (2)$$

From (1) and (2), we get (0).

3. (a) $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, $\mu(A_n) = \infty$, $\bigcap_{n \geq 1} A_n = \emptyset$. So $\mu(\bigcap_{n \geq 1} A_n) \neq \lim_{n \rightarrow \infty} \mu(A_n)$. Hence m.c.f.a doesn't hold.
- (b) Take $\mu'(A) = \sum_{i \in A} \frac{1}{2^i}$ (or we can choose any discrete probability measure). Then $\mu(\bigcap_{n \geq 1} A_n) = 0 = \lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} = \lim_{n \rightarrow \infty} \mu(A_n)$. So m.c.f.a holds.
- (c) If $\exists n_0$ s.t $\mu(A_{n_0}) < \infty$, then m.c.f.a holds.
- (d) Always true.

4. (a) Let $\mathcal{C} = \{(a, b] : -\infty \leq a < b < \infty\} \cup \{(a, \infty), a \in \mathbb{R}\}$. It is a semi-algebra. Define

$$\begin{aligned} \mu_F(a, b] &= F(b+) - F(a+) \\ &= (1 - e^{-b}) - (1 - e^{-a}) \\ &= e^{-a} - e^{-b} \\ \mu_F(a, \infty] &= F(\infty-) - F(a+) \\ &= 1 - (1 - e^{-a}) \\ &= e^{-a} \end{aligned}$$

Then this defines a measure on semi-algebra $\mathcal{C}$.

By Caratheodory Extension Theorem, there exists an extension of μ_F , called μ_F^* on $(\Omega, \mathcal{M}_{\mu_F^*}, \mu_F^*)$, where $\mathcal{M}_{\mu_F^*}$ = set of all μ_F^* -measurable sets on $\Omega = \{A \in \mathcal{P}(\Omega) : \mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^C), \forall E \in \Omega\}$. $\mathcal{M}_{\mu_F^*}$ is a σ -algebra containing $\mathcal{C} \Rightarrow \mathcal{B}(\mathbb{R}) \subset \sigma(\mathcal{C}) \subset \mathcal{M}_{\mu_F^*}$. μ_F^* restricted to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an extension of μ_F .

- (b) This is unique since $\mu_F^*(\mathbb{R}) = 1$ (finite measure, hence any other measure that takes same value on $\mathcal{C}$ will take same values on $\mathcal{B}(\mathbb{R})$).
- (c) $A = (-\infty, -2) \cup (2, \infty)$. So, $\mu_F^*(A) = \mu_F^*((2, \infty)) = e^{-2}$, since $\mu_F^*((-\infty, -2)) = 0$.